Homework #6

This is the homework for week 7. The due date is 11:59 p.m., October 27. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1):** (20 points) Ledolter and Hogg report that two rubber compounds were tested for tensile strength. Rectangular materials were prepared and pulled in a longitudinal direction. A sample of 14 specimens, seven from compound A and seven from compound B, was prepared, but it was later found that two B specimens were defective and they had to be removed from the test. The tensile strength (in units of 100 pounds per square inch) of the remaining specimens are as follows:

$$A: \ 32\ 30\ 33\ 32\ 29\ 34\ 32, \ B: \ 33\ 35\ 36\ 37\ 35.$$

i. Calculate an approximate 95% two–side confidence interval for the difference of the mean tensile strengths of the two rubber compounds $\mu_A - \mu_B$. You may assume that the data satisfy normal distributions.

ii. Derive the upper–side confidence interval for $\mu_A - \mu_B$.

i. From the central limit theorem,

$$\frac{\widehat{\mu}_A - \widehat{\mu}_B}{\sqrt{\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}}} \to_d N(0,1).$$

Since

$$\widehat{\mu}_A = 31.71, \ \widehat{\mu}_B = 35.20, \ S_A^2 = 2.49, \ S_b^2 = 1.76,$$

the 95% confidence interval is given by

$$\mu_A - \mu_B \in [-5.15, -1.82].$$

ii. The 95% upper-side CI is given by

$$\mu_A - \mu_B \in (-\infty, -2.11].$$

(**Question2):** (15 points) In developing countries in Africa and the Americas, let p_1 and p_2 be the respective proportions of women with nutritional anemia. Find an approximate 90% two–side confidence interval for $p_1 - p_2$, given that a random sample of $n_1 = 2100$ African women yielded $y_1 = 840$ with nutritional anemia and a random sample of $n_2 = 1900$ women from the Americas yielded $y_2 = 323$ women with nutritional anemia.

Answer: The ratio is $\widehat{p}_1 = \frac{840}{2100} = 0.4$ and $\widehat{p}_2 = 323/1900 = 0.17$. From central limit theorem,

$$\frac{\widehat{p}_1 - \widehat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{\widehat{p}_1(1-\widehat{p}_1)}{n_1} + \frac{\widehat{p}_2(1-\widehat{p}_2)}{n_2}}} \to_d N(0,1),$$

so the 90% confidence interval is given by

$$p_1 - p_2 \in [0.21, 0.25]$$

(**Question3):** (45 points)

Suppose data $X_1, X_2, \cdots, X_n$ are i.i.d., satisfying normal distribution $N(\mu, \sigma^2)$, where σ^2 is known. After collecting data, the statistician wants to test the hypothesis:

$$H_0 : \mu = 0,\ H_1 : \mu = 1$$

with Type–I error α.

i. Derive the test statistics and the rejection of the test. Notably, the answer may not be unique, and as long as it is plausible, it should be fine.

ii. Calculate Type–I error of your test, ensure it to be α.

iii. Calculate Type–II error of your test, and prove that the Type–II error tends to 0 as $n \to \infty$.

iv. Suppose $\sigma^2 = 1$ and $\alpha = 5\%$. The data we collects is

[-1.78, -0.53, -0.40, 0.15, -0.31, -1.07, -1.32, -0.80, -0.89, -0.40, 0.10, 0.67, -1.25, -1.29, 0.04]. Based on this observation, which hypothesis do you prefer.

i. We choose the test statistics and the rejection region

$$\widehat{T} = \frac{\sqrt{n}\widehat{\mu}}{\sigma}, R_\alpha = [z_\alpha, \infty).$$

ii. Under H_0,

$$Prob(\widehat{T} \in R_\alpha) = Prob(\widehat{T} \geq z_\alpha) = \alpha.$$

iii. Under H_1,

$$\widehat{T} = \frac{\sqrt{n}(\widehat{\mu} - 1)}{\sigma} + \frac{\sqrt{n}}{\sigma},$$

so

$$Prob(\widehat{T} \notin R_\alpha) = Prob(\frac{\sqrt{n}(\widehat{\mu} - 1)}{\sigma} \leq z_\alpha - \frac{\sqrt{n}}{\sigma}) = \Phi\left(z_\alpha - \frac{\sqrt{n}}{\sigma}\right),$$

which tends to 0 as $n \to \infty$.

iv. The sample mean is $\widehat{\mu} = -0.61$ and $\widehat{T} = -2.34 < 1.64$. Thus, we choose H_0.

2 Programming Questions

(20 points)

Implement the following experiment using Jupyter or R:

i. Generate $X_1, \cdots, X_n$ satisfying normal distribution $N(0, 1)$.

ii. Suppose we want to test the hypothesis

$$H_0 : \mu = 0 \text{ and } H_1 : \mu = 1.$$

Perform test using the test statistics and the rejection region derived from Question 3. Suppose $\sigma^2 = 1$ and $\alpha = 5\%$ is known.

Repeat Step i and ii for 200 times, record the frequency of rejecting H_0. It should be closed to 5%.

iv. Now redo step i to iii for distribution $N(1, 1)$. Now the frequency or rejecting H_0 should be significantly larger than 5%.